

Duplicate Resolution + Land-Cover Bias Audit

Generated 2026-09-02, before any model training — this is exactly the kind of check requested to happen before trusting a "great result" tomorrow. No model has been trained. Both scripts referenced here are read-only investigations; `build_training_dataset.py` was re-run to apply the duplicate fix, nothing else changed.

Part 1 — Duplicate resolution

Investigation: same event or genuinely distinct?

All 3 duplicate-coordinate pairs were pulled with their full source rows (`scripts/ml/resolve_duplicate_positives.py, run_investigation()`):

Pair 1 — (27.393056, 88.635278), "Kabi Landslide" | | Slide_No | History | |--|---|---| | Row A | .../2018/serial no. (incomplete) | Mid-July 2018, 2 August 2018 | | Row B | .../2020/03 | Mid-July 2018, 2 August 2018, **24 May 2020** |

Same name, same location description ("State Highway; 4 km before Kabi village"), same material/movement. Row B's history is a strict superset of Row A's — this is GSI re-logging the **same recurring failure site** after it slid again in 2020. **Clear duplicate**, not two events.

Pair 2 — (27.426361, 88.583028) Both rows: same year (2015), sequential serial numbers (.../74, .../75), identical district/location/material/movement, no name or history on either to differentiate, coordinates identical to 6 decimal places. **Ambiguous** — could be a duplicate entry, or two closely-spaced slides logged against the same rounded field GPS waypoint. No evidence in the record distinguishes them.

Pair 3 — (27.437889, 88.601694), "Chawang" Same pattern as Pair 2 (2015, sequential serials .../07/.../08), one row has the place name "Chawang," the other doesn't. Same ambiguity.

Resolution and why it's defensible either way

All three treated as duplicates and deduplicated. For Pair 1 this is close to certain. For Pairs 2 and 3, it's a judgment call — but a low-risk one: two rows at the **exact same coordinate** produce **identical DEM and land-cover feature values** (same pixel). Whether they're truly one event or two, keeping both adds zero new information and only inflates that one location's weight in training. Deduplicating costs nothing either way.

Rows removed (kept the more complete record — has a name, or the longer history — per pair): | Dropped Slide_No | Coordinate | Reason | |--|---|---| | State: Sikkim/district: North.../2018/serial no. | 27.393056, 88.635278 | Superseded by the 2020-updated record for the same site | | SKM/NS/78A11/2015/75 | 27.426361, 88.583028 | No distinguishing info vs. .../74; lower serial kept | | SKM/NS/78A11/2015/08 | 27.437889, 88.601694 | No name vs. .../07, which has "Chawang" |

777 → 774 positives. Wired into `build_training_dataset.py` so this is now automatic on every rebuild, not a one-off manual fix.

Re-run integrity checks (post-dedup)

Check	Result
Exact duplicate coordinates	0 (was 6)
Positive/negative coordinate overlap	0
Min inter-class distance	203.7m (≥ 200m exclusion buffer)

Class balance	774 / 774, ratio 1.000
District proportions	Exact match, all districts
Missing / infinite values	None
Leakage-prone fields	None present

Final dataset: **1548 rows** (774 + 774), same 9 columns as before (path unchanged: `data/processed/training_dataset.csv`).

Part 2 — Land-cover bias investigation

The question

The earlier review found built-up land cover at 9.0% of positives vs 0.3% of negatives (~30×). Before trusting that as genuine terrain signal, checked whether it's actually a restatement of the known road-survey bias (positives are GSI road-cut inspections — median distance to road is tiny; negatives are spread across a wider corridor).

Important: `distance_to_road` was computed here **purely as a diagnostic**. It is not, and will not be, added to `training_dataset.csv` or used as a model feature — using it as a feature would leak the sampling process itself into the model. Using it here to *check* for that leakage is the opposite move.

What the numbers show

Distance-to-road is a strong confound in this dataset, as expected: point-biserial correlation between label and distance-to-road is **r = -0.545** (positives median 9.6m from a road; negatives median 127m). This is exactly why distance-to-road is correctly excluded as a feature — it would be close to the strongest "predictor" available, and it would just be re-encoding how positives were surveyed.

Built-up land itself clusters near roads, independent of label: median distance-to-road for built_up pixels (both classes combined) is 8.7m, vs. 29.6m for tree_cover. This is an expected, real geographic pattern — settlements and junctions sit on roads.

The critical test — does the gap survive controlling for distance? Stratifying by distance-to-road bin:

Distance to road	Positives built-up %	Negatives built-up %
0–20m	10.6% (67/633)	0.0% (0/86)
20–50m	3.4% (3/89)	1.0% (1/98)
50–150m	0.0% (0/28)	0.4% (1/258)
150m+	~0%	~0%

If built-up were *purely* a road-proximity proxy, this gap should shrink to near-zero once distance is held constant — instead, at the same 0–20m proximity, positives are built-up **10.6% of the time and negatives 0%**. See `data/processed/landcover_bias_audit.png` — the bars don't converge.

- Is 0/86 just a small sample?** No — checked. If negatives near roads truly had built-up rate similar to positives' 10.6%, the probability of observing zero out of 86 by chance is roughly 1 in 10,000. This gap is real, not noise.

Honest interpretation

Built-up land cover carries **signal beyond simple road proximity** — the confounding hypothesis ("it's just road distance in disguise") does not fully hold up. But two mechanisms remain equally plausible and this data can't separate them: - **Genuine anthropogenic destabilization**: construction/road-cutting near settlements measurably weakens slopes — a real, literature-supported effect - **Reporting/documentation bias**: GSI may be more likely to *log* a landslide that threatens a building or blocks a road near a settlement than one that fails unreported in remote forest — a bias in what gets into the inventory at all, not in the terrain itself

Either way, built-up land's association with landslides in this dataset is not simply the road-survey artifact restated. It's a real pattern in the data, with an honestly ambiguous cause.

Two candidate feature sets, ready for approval

Set	Features
Baseline	elevation, slope, curvature, distance_to_drainage
Extended	Baseline + land_cover_class (one-hot)

Both are subsets of columns already in `data/processed/training_dataset.csv` — no rebuild needed to select either at training time. Per instruction, both will be trained and evaluated on the **same spatially defensible holdout** (not a naive random split, given how spatially clustered this data is) once you approve moving forward.

Not yet done: training either model. Waiting for the go-ahead.